EDUARDO LUIS

ELECTRICAL & ELECTRONICS ENGINEER | EMBEDDED SYSTEMS | CONTROLS & INSTRUMENTATION

3225 Woodland Park Dr, Houston, TX, 77082 • 832-430-0514 • zaului4@gmail.com

SUMMARY

Electrical and Electronics Engineer with a B.S. in Electronic Systems Engineering Technology from Texas A&M University and hands-on experience in industrial electrical systems, motor controls, electrical testing, controls, instrumentation, embedded systems, and hardware validation. Experience supporting industrial transformers, 380 V and 480 V three-phase power systems, AC and DC motors, contactors, relays, breakers, sensors, encoders, motor drives, PLC I/O, and control circuits. Experienced analyzing electrical schematics, troubleshooting field wiring and control systems, and performing electrical testing using multimeters, Megger testers, and phase-sequence equipment. Strong foundation in circuit analysis, control systems, PCB design, embedded systems, sensors, motors, and automated testing.

RELEVANT PROJECTS

Senior Capstone Project , EcoTrace Wildlife Tracking System **Agrilife, 2025 - 2026**

- Designed and integrated a low-power embedded tracking system for monitoring Eastern fox squirrels, combining an nRF52840 MCU, GNSS, BMI270 accelerometer, SPI flash memory, LiPo battery, and BLE communication.
- Developed the electrical system architecture and selected components based on power, size, communication, environmental, and reliability requirements.
- Designed schematics, PCB layouts, and functional block diagrams in Altium Designer, translating system requirements into hardware implementation.
- Implemented UART, I2C, SPI, and BLE interfaces, and built a store-and-forward architecture for GPS acquisition, local storage, and wireless transfer to an ESP32-C3 gateway.
- Performed hardware integration, testing, troubleshooting, and system-level debugging, collaborating with multidisciplinary teams to balance cost, power, size, and reliability.

Heater & Automatic Stirring System **Texas A&M University, 2023 - 2025**

- Designed and developed an automated electrical system integrating a heating element, Peltier cell, temperature sensor, and DC motor, using closed-loop sensor feedback and proportional control to regulate temperature toward a setpoint from wall power.
- Modeled the thermal and motor subsystems using plant transfer functions, controller gains, and feedback loops in **MATLAB/Simulink**, analyzing rise time, stability, and system response to evaluate thermal-motor interaction.
- Troubleshot sensor, thermal, Peltier, and motor performance issues, collaborating with a multidisciplinary engineering team to develop system architecture, divide responsibilities, and document the system.

RELEVANT WORK EXPERIENCE

Electrical Engineer @ Jindal Pipe USA – Baytown, TX | **June 2026 – Present**

- Work with industrial transformers, 380 V/480 V three-phase systems, three-phase AC motors, and DC motors.
- Analyze electrical schematics, motor control circuits, ladder logic, PLC I/O, and control wiring for industrial equipment.
- Troubleshoot motors, contactors, relays, breakers, sensors, encoders, VFDs, and motor control systems.
- Gain hands-on experience with Allen-Bradley PowerFlex drives, Siemens controls, servo drives, servo motors, and PLC-controlled systems.
- Perform electrical testing using digital multimeters, Megger insulation testers, and phase-sequence testing equipment.
- Trace sensor and control wiring through junction boxes and panels; use SAP for maintenance material reservations and issue slips.
- Supervised maintenance on motors, motor cables, cable tracks, hoses, sensors, and electrical panel components, following plant safety and electrical work procedures.

Semiconductor Test & Electrical Validation

- Developed electrical test procedures for semiconductor and electronic devices using **Teradyne ETS-364/88** and **Eagle test** environments, performing functional and parametric testing of voltage, current, resistance, leakage, timing, and digital functionality
- Built automated **LabVIEW/TestStand** workflows to analyze results and troubleshoot performance.

Circuit & Electrical Systems Design

- Designed and analyzed AC/DC circuits using mesh, nodal, and transient analysis methods. Analyzed transmission lines, RF components, and impedance matching in high-frequency systems.
- Designed analog/digital and power-supply circuits with diodes, BJTs, MOSFETs, and op-amps. Developed schematics and simulations in Altium Designer and Multisim, selecting components.

Microcontroller-Based Embedded Electrical Systems

- Developed embedded systems on the TI MSP432P401R using ARM assembly and register-level code.
- Programmed GPIO, timers, ADC, and UART interfaces, debugging registers and memory behavior.

EDUCATION

Texas A&M University – College Station, TX | **Graduated**

Bachelor of Science in Electronic Systems Engineering Technology

Minors: Cybersecurity, Engineering Project ManagementSpecialization in Advanced Manufacturing.

Houston Community College – Houston, TX | Graduated

Associate in Electrical Engineering (Honors Student) GPA: 3.8 Honors

Design and Project Management.

- Phi Theta Kappa Honor Society
- National Society of Collegiate Scholars (NSCS)
- **Languages:** English, Portuguese, Spanish, French
- **Certifications:** Lean Six Sigma Yellow Belt

Technical Skills: KiCad, Altium Designer, OrCAD Capture, Allegro PCB Designer, AutoCAD, Multisim, Eagle Test Systems, Python, C, Swift, MATLAB, MQL4, MQL5, LabVIEW, TestStand, Oscilloscopes, Hardware Validation, Semiconductor Testing, Automated Testing, Circuit Debugging, Calibration, Mixed-Signal Testing, Wireshark, VPN Configuration, ACLs, Firewalls, Packet Analysis, PLC